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Accelerating the discovery of insensitive high-energy-density materials by a materials genome approach

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Finding new high-energy-density materials with desired properties has been intensely-pursued in recent decades. However, the contradictory relationship between high energy and low mechanical sensitivity makes the innovation of insensitive high-energy-density materials an enormous challenge. Here, we show how a materials genome approach can be used to accelerate the discovery of new insensitive high-energy explosives by identification of "genetic" features, rapid molecular design, and screening, as well as experimental synthesis of a target molecule, 2,4,6-triamino-5-nitropyrimidine-1,3-dioxide. This as-synthesized energetic compound exhibits a graphite-like layered crystal structure with a high measured density of 1.95 g cm^{-3}, high thermal decomposition temperature of 284 °C, high detonation velocity of 9169 m s^{-1}, and extremely low mechanical sensitivities (impact sensitivity, >60 J and friction sensitivity, >360 N). Besides the considered system of six-member aromatic and hetero-aromatic rings, this materials genome approach can also be applicable to the development of new high-performing energetic materials.

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Since the invention of black powder in China and the era of nitroglycerin brought by Alfred Nobel, high explosives have contributed enormously to the progress and prosperity of mankind[1,2]. With the rapid development of modern high-energy-density materials (HEDMs), a large variety of high-performing energetic materials have been developed and continue to play an overwhelming role in some of the today's fantastic engineering projects and space exploration[3–19]. However, the increasing severity of safety issues in both military and civil applications renders the menu of available insensitive high explosive (IHE) molecules vanishingly small[20,21]. To date, the only approved IHE molecule that meets the criteria of United States Department of Energy is 1,3,5-triamino-2,4,6-trinitrobenzene (TATB)[22], a high-density aromatic compound first synthesized in 1888[23]. Owing to the extremely low sensitivities to external stimuli, TATB is also described as "wood explosive"[24,25] and usually preferred for applications where extreme safety is required, such as the explosive formulations in nuclear weapons and low burning rate propellant components for space exploration. However, the detonation energy of TATB is relatively low, around 65% of the widely used high explosive 1,3,5,7-tetranitro-1,3,5,7-tetrazocane (HMX)[26]. Over the past decades, a few low-sensitivity HEDMs, such as 1,1-diamino-2,2-dinitroethene(FOX-7) and 2,6-diamino-3,5-dinitropyrazine-1-oxide (LLM-105), have been reported as potential TATB replacements, but their safety issues are still far away from industrial standard of IHE molecules[27–29]. Ideally, new IHE molecules are expected to have the TATB's sensitivity and the HMX's energy. But in most cases, the high detonation performances of HEDMs are accompanied with the undesirable high sensitivity to external stimuli such as impact, shock, spark, friction, and heat. This contradictory relationship between high energy and low sensitivity makes the development of new high-performing IHE molecules a long-standing challenge in chemical sciences[30].

As compared to the revolutionary advances in other functional materials, the progressive process of new insensitive HEDMs is painfully slow, probably due to that the traditional synthesis of high explosives still relies primarily on scientific intuition and trial-and-error[31,32]. In 2011, the US government launched the Materials Genome Initiative (MGI), which aims to reduce the development time by providing the infrastructure and training that innovators need to discover new advanced materials in a more efficient and economical way. In recent years, the materials genome approach has been applied in new material innovations and helped scientists successfully discover new alloy materials[33,34], metal-organic frameworks (MOF)[35,36], and inorganic porous materials[37,38]. These fascinating advances suggest the success of materials genome approach for the new materials discovery.

Here, we show how the materials genome approach has been used for the rapid design and structural screening of new IHE molecules with desired properties, as this allows us to quickly choose potential IHE molecule candidates for the experimental synthesis. In this approach, we first identify three key "genetic" features of IHE molecules from large amounts of component and structural information. The second step involves the rapid design and screening of candidate molecules with the guidance of identified "genetic" features. Third, we experimentally synthesized the screened target molecule. The property studies showed that the as-synthesized energetic compound (also named as ICM-102) was a promising IHE molecule with a high measured density (1.95 g cm^{-3}), high thermal decomposition temperature (284 °C), high detonation velocity (9169 m s^{-1}), and extremely low sensitivities (like TATB) to external stimuli such as impact, friction, and electrostatic spark.

Results

Identification of key "genetic" features of IHE molecules. Before the computationally-guided design and rapid screening of new IHE molecules by a materials genome approach, the first challenge in this work is the identification of key material "genetic" features from those reported IHE molecules. Since high density and low sensitivity are the two most important properties for high-performing IHE molecules, we attempt to explore the "genetic code" of typical HEDMs, i.e., to reveal the intrinsic relationships between materials' structure (e.g., molecular constitution and crystal packing) and properties (e.g., high density and low sensitivities) from large bodies of experimental data. It is well known that four basic chemical bases (A, T, G, C) provide the super-complex genetic information of organism by some specific arranged sequences and intermolecular interactions (Fig. 1). An extraordinary coincidence is that most HEDMs are generally composed of four basic elements (C, H, N, O) and their

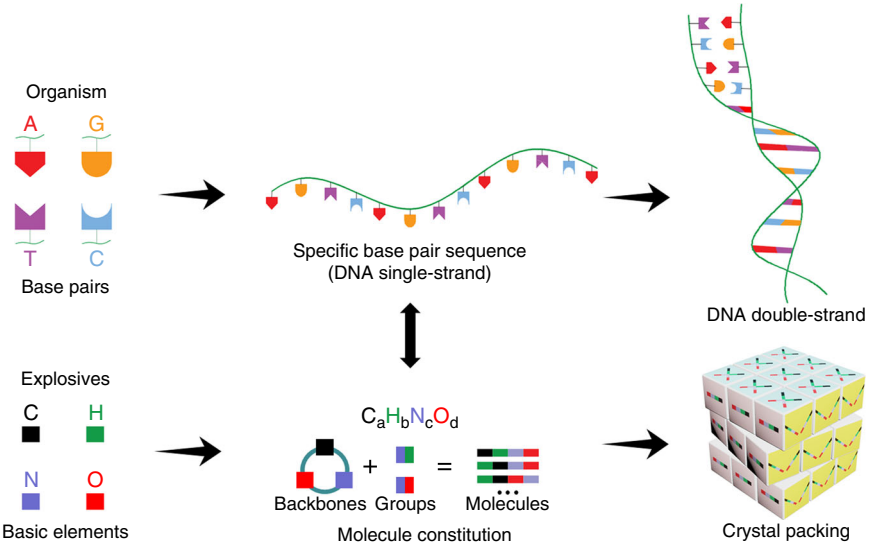


Fig. 1 Illustration of possible materials genomes of organic explosives. Comparison of materials genome approach for organic explosives with that of an organism